

Cover letter for Nature Energy

Draft for author review — not cleared for submission | 27 September 2026

Dear Editors,

We are preparing a substantially revised manuscript entitled “Service constraints and power boundaries shape the grid value of flexible AI computing”. This draft relates to manuscript NENERGY-26093649; the appropriate route for any revision must follow the journal's actual decision and instructions.

The study examines the conditions under which changes in AI operating mode and task timing can be interpreted as energy-system flexibility. A controlled four-policy comparison holds the task cohort and completion conditions fixed. Across 108 conditional GPU-model configurations, the attribution depends strongly on the allowed completion window; lower modeled energy use can coexist with higher hourly GPU demand. Analytical counterexamples show how cohort double counting can reverse a capacity conclusion and how linear procurement-response extrema can miss nonlinear commitment costs.

We believe that a fully validated version of this work could interest Nature Energy readers by clarifying the evidence needed to translate computing flexibility into power-system value. The present findings are conditional modeling and source-audit results. They do not yet demonstrate measured paired whole-node energy savings, production-SLA compliance or validated provincial capacity benefits. The revision must obtain these missing measurements and physical inputs and complete fixed-investment full-year evaluation before a submission-ready claim can be made.

This letter will be finalized only after the empirical results, uncertainty, reproducibility materials and final author declarations are complete. All-author approval, competing interests, related submissions and the status of the existing manuscript require confirmation; no declaration is made here on the authors' behalf.

Sincerely,

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